Lattice Theory (2)

CSE552 Program Analysis — Lecture 6

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Motivating Example — Sign^6

```
a = 42;
b = a + input();
```

Abstract domain: Sign^6

- $a_0 = \top$
- $b_0 = \top$
- $a_1 = +$
- $b_1 = b_0$
- $a_2 = a_1$
- $b_2 = a_1 + \top$
  (we will define $+$ later)

Motivating Example — (Var $\to$ Sign)^3

```
a = 42;
b = a + input();
```

Abstract domain: (Var $\to$ Sign)^3

- $x_0 = [a \mapsto \top,\ b \mapsto \top]$
- $x_1 = x_0[a \mapsto +]$
- $x_2 = x_1[b \mapsto x_1(a) + \top]$

Flow-Sensitive Analysis

```
a = 42;
b = a + input();
```

Abstract domain: $(\mathrm{Var} \to \mathrm{Sign})^3$

- $x_0 = [a \mapsto \top,\ b \mapsto \top]$
- $x_1 = x_0[a \mapsto +]$
- $x_2 = x_1[b \mapsto x_1(a) + \top]$

- Flow-sensitive analysis
  - The order of statements is taken into account
  - The sign of variables is determined for each program point

Solving by Substitution

```
a = 42;
b = a + input();
```

Abstract domain: $(\mathrm{Var} \to \mathrm{Sign})^3$

- $x_0 = [a \mapsto \top,\ b \mapsto \top]$
- $x_1 = x_0[a \mapsto +]$
- $x_2 = x_1[b \mapsto x_1(a) + \top]$

- For this example program, each equation only depends on preceding ones
  - The solution can be found by simple substitution
  - $x_0 = [a \mapsto \top,\ b \mapsto \top]$
  - $x_1 = [a \mapsto +,\ b \mapsto \top]$
  - $x_2 = [a \mapsto +,\ b \mapsto \top]$
- In general, mutually recursive equations may appear, e.g., for programs that contain loops

Fixed Point Formulation

- Solving this system requires finding the fixed point for function $f : (\mathrm{Var} \to \mathrm{Sign})^3 \to (\mathrm{Var} \to \mathrm{Sign})^3$ defined as follows:
  - $f(x_0, x_1, x_2) = ([a \mapsto \top, b \mapsto \top],\ x_0[a \mapsto +],\ x_1[b \mapsto x_1(a) + \top])$
  - A fixed point for f is x that satisfies $f(x) = x$
- How can we find a fixed point for a function over a lattice?

Monotone Functions — Definition

Definition (Monotone function). A function $f : L_1 \rightarrow L_2$ where L_1 and L_2 are lattices is monotone (or order-preserving) when

$$\forall x, y \in L_1.\ x \sqsubseteq y \Rightarrow f(x) \sqsubseteq f(y)$$

- From the analysis perspective, the intuition of monotonicity is that more precise input does not result in less precise output

Extensive and Distributive Functions

Definition (Extensive function). A function $f : L \rightarrow L$ where L is a lattice is extensive when $\forall x \in L.\ x \sqsubseteq f(x)$

Definition (Distributive function). A function $f : L_1 \rightarrow L_2$ where L_1 and L_2 are lattices is distributive when

$$\forall x, y \in L_1.\ f(x) \sqcup f(y) = f(x \sqcup y)$$

- Every distributive function is also monotone
- Not every monotone function is also distributive

Monotone Functions — Properties (1)

Important properties:

- Every constant function is monotone
- f is monotone $\iff \forall x, y.\ f(x) \sqcup f(y) \sqsubseteq f(x \sqcup y)$
- If f and g are monotone, then so is their composition $g \circ f$, which is defined by $(g \circ f)(x) = g(f(x))$
- $\sqcup : L^2 \to L$ and $\sqcap : L^2 \to L$ are monotone

Monotone Functions — Properties (2)

Important properties (cont.):

- If $f : L_1 \to (A \to L_2)$ and $g : L_1 \to L_2$ are monotone, then so is the function $h : L_1 \to (A \to L_2)$ defined by $h(x) = f(x)[a \mapsto g(x)]$
- $f_1 : L \to L_1, \ldots, f_n : L \to L_n$ are monotone $\iff$ $f : L \to L_1 \times \cdots \times L_n$ defined by $f(x) = (f_1(x), \ldots, f_n(x))$ is monotone

Monotone Functions — Example

- $f(x_0, x_1, x_2) = ([a \mapsto \top, b \mapsto \top],\ x_0[a \mapsto +],\ x_1[b \mapsto x_1(a) + \top])$

$$= (f_0(x_0, x_1, x_2),\ f_1(x_0, x_1, x_2),\ f_2(x_0, x_1, x_2))$$

- f_0 is monotone because it is a constant function
- f_1 is monotone because $(x_0, x_1, x_2) \mapsto x_0$ is monotone and $(x_0, x_1, x_2) \mapsto +$ is monotone
- f_2 is monotone (we will show it later)
- f is monotone because f_0, f_1, and f_2 are monotone

Fixed Points — Definition

Definition (Fixed point).

- $x \in L$ is a fixed point for f if $f(x) = x$
- A least fixed point (lfp) x for f is a fixed point for f where $x \sqsubseteq y$ for every fixed point y for f

Fixed Points — Role in Analysis

Where the constraints are expressed as an equation system $x = f(x)$,

- A solution to the system is the same as a fixed point for f
- For carefully designed constraints, every fixed point provides a sound result
- Among all fixed points, the lfp provides the most precise result

Tarski's Fixed Point Theorem

Theorem (Tarski[1]). If L is a complete lattice and $f : L \to L$ is monotone, then f has a least fixed point

- The most precise solution is guaranteed to exist, but how can we find it?

[1] A lattice-theoretical fixpoint theorem and its applications (Tarski, 1955)

Kleene's Fixed Point Theorem

Theorem (Kleene[2]). If L is a complete lattice with a finite height and $f : L \to L$ is monotone, then

$$\mathrm{lfp}(f) = \bigsqcup_{i \geq 0} f^i(\bot)$$

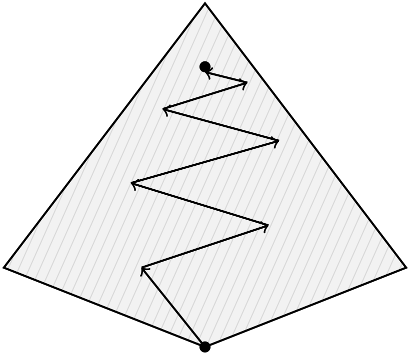
- If the lattice has a finite height, we can find the lfp by computing the increasing chain $\bot \sqsubseteq f(\bot) \sqsubseteq f^2(\bot) \sqsubseteq \cdots$ until the fixed point is reached

[2] Introduction to metamathematics (Kleene, 1952)

Naive Fixed Point Algorithm

NaiveFixedPointAlgorithm(f)**:**

$x \leftarrow \bot$
while $x \neq f(x)$ **:**
  $x \leftarrow f(x)$
return x



Fixed Point Algorithm — Example

- $f(x_0, x_1, x_2) = ([a \mapsto \top, b \mapsto \top],\ x_0[a \mapsto +],\ x_1[b \mapsto x_1(a) + \top])$
- $\bot = ([a \mapsto \bot, b \mapsto \bot],\ [a \mapsto \bot, b \mapsto \bot],\ [a \mapsto \bot, b \mapsto \bot])$
- $f(\bot) = ([a \mapsto \top, b \mapsto \top],\ [a \mapsto +, b \mapsto \bot],\ [a \mapsto \bot, b \mapsto \bot])$
- $f^2(\bot) = ([a \mapsto \top, b \mapsto \top],\ [a \mapsto +, b \mapsto \top],\ [a \mapsto +, b \mapsto \top])$
- $f^3(\bot) = ([a \mapsto \top, b \mapsto \top],\ [a \mapsto +, b \mapsto \top],\ [a \mapsto +, b \mapsto \top]) = f^2(\bot)$

Precision and Complexity

- Even though we find the most precise possible solution to the equation system, the equation system is merely a conservative approximation of the actual program behavior
- The semantically most precise answer can be below the lfp in the lattice
- The time complexity of computing a fixed point with this algorithm depends on
  - The height of the lattice because this provides a bound for the number of iterations of the algorithm
  - The cost of computing $f(x)$ and testing equality, which are performed in each iteration

Inequality Constraints

- Some analyses can yield inequations
- We can rewrite them as equations
  - $x \sqsupseteq f(x)$ is equivalent to $x = x \sqcup f(x)$
  - $x \sqsubseteq f(x)$ is equivalent to $x = x \sqcap f(x)$

Summary

- Solving constraints can be formulated as finding a fixed point for a function over a lattice
- Monotone functions on complete lattices have a least fixed point
- The naive fixed point algorithm iterates f from $\perp$ until convergence